

Understanding DAO Governance: Proposal Categorization and its Impact on Online Voter Behavior

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Abstract—Decentralized Autonomous Organizations (DAOs) have emerged as a new paradigm for online governance, enabling large online communities to make collective decisions through blockchain-based voting systems. While prior work has examined DAO structures, centralization, and participation rates, the role of proposal content in shaping governance dynamics remains poorly understood. We present a large-scale, cross-platform analysis of over 32,000 proposals from four major DAO ecosystems: Snapshot, Service Nervous System (SNS), OpenGov, and DAOstack. We develop a proposal categorization framework that combines transformer-based embeddings, dimensionality reduction, and clustering, and find that proposal topic is significantly associated with approval outcomes on every platform (Cramér’s V 0.20–0.56). Across platforms, financial and treasury proposals dominate activity and take the longest to resolve, while technical and operational proposals are typically shorter and more frequently approved. The topics that face the highest rejection rates differ by platform. We further identify substantial cross-platform differences in the mix of proposal topics, proposer concentration, and the temporal evolution of governance priorities. These findings offer new insights for designing more transparent, efficient, and inclusive decentralized governance systems.

I. INTRODUCTION

DAOs enable large online communities to coordinate and govern themselves through blockchain-based voting systems. The DAO ecosystem has grown rapidly in recent years, exceeding \$22.8B in treasury value and involving millions of active voters [1]. DAOs today govern a wide range of applications, including decentralized finance (DeFi), social media, AI and gaming. As these systems gained economic relevance, prior work studied governance structures, voter participation, and centralization risks [2], [3], [4], [5], [6], [7], [8], [9], [10].

DAO governance typically revolves around proposals submitted and voted on by community members [11], [12]. Depending on the platform, governance can be executed either on-chain through smart contracts or off-chain via external voting systems. While on-chain governance provides transparency and automation, it may suffer from participation barriers such as transaction costs. Off-chain governance improves accessibility and scalability, but introduces additional trust assumptions [13], [14]. Across both governance models, voting power concentration and participation inequality remain important challenges [10], [15], [16].

Despite increasing research interest in DAO governance, one important dimension remains underexplored: the content of the proposals themselves. Proposal topics, such as treasury allocations, technical upgrades, or community initiatives, likely shape voter behavior, participation, and approval dynamics. However, a systematic cross-platform analysis of proposal content and its relationship to governance outcomes is still missing. This paper addresses the following research question:

- How can DAO proposals be effectively categorized, and how do these categories influence governance dynamics and voter behavior across different DAO ecosystems?

Understanding proposal categories can help identify community priorities, reveal governance patterns, and improve decision-making processes. In particular, proposal categorization enables analyzing which topics dominate governance activity, which proposal types are more likely to fail, and how governance priorities evolve over time.

Our Contributions

This paper presents a large-scale analysis of proposal content and governance dynamics across multiple ecosystems.

Dataset: We collect and standardize a dataset of more than 32,000 proposals from four major DAO platforms: Snapshot, DAOstack, Service Nervous System, and OpenGov. To support reproducibility and future research, we release all data and the data-collection code as open source ¹.

Categorization framework: We develop a proposal-level categorization pipeline combining transformer-based embeddings, PCA-based dimensionality reduction, BERTopic, and k-means clustering. We additionally compare the resulting topics against an LDA-based baseline.

Cross-platform governance analysis: Using the extracted categories, we analyze proposal distributions, approval and rejection rates, proposer diversity, proposal length, and temporal governance trends across platforms.

Key findings: Our results show that financial and treasury-related proposals dominate governance activity across platforms and take the longest to resolve, while technical and operational proposals are typically shorter and more frequently approved. The topics with the highest rejection rates differ by

¹<https://zenodo.org/records/17280115>

platform (financial on SNS; community and creative on Snapshot; events and flawed submissions on OpenGov; speculative token-generation on DAOstack). We further observe substantial cross-platform differences in proposer concentration, the mix of proposal topics, and governance dynamics.

II. DAO FRAMEWORKS AND DATA COLLECTION

In this section we describe the four DAO platforms studied in this paper, highlighting the governance methods they offer and how we collected the proposal dataset. Table I contains an overview of the collected data from different platforms. In total, 32,819 proposals from 144 different DAOs across four platforms were collected for the NLP process described in the subsequent section. Similarly to related work, we focus on active DAOs with a minimum number of proposals and voters. We chose these four platforms deliberately to span the design space rather than to exhaustively enumerate it: Snapshot represents large-scale off-chain governance, SNS and OpenGov represent two contrasting on-chain models (per-DAO smart contracts vs. a single network-wide governance system), and DAOstack provides a now-discontinued reputation-weighted point of comparison. The minimum activity thresholds applied to each platform (detailed below) are aimed at excluding DAOs whose proposal volume or voter participation is too low to support topic modeling or to reflect a community-driven outcome relative to more active DAOs on the same platform.

Platform	# DAOs	# Proposals	Time Span
Snapshot (off-chain)	59	18,702	2020-2025
SNS	27	9,058	2023-2025
OpenGov	1	1,486	2023-2025
DAOstack (inactive)	57	3,573	2019-2023
Total	144	32,819	

TABLE I: Overview of DAO platforms and proposal counts.

a) Snapshot: Snapshot² [17] is an open-source off-chain governance platform widely used by DeFi protocols and NFT communities; votes are not executed on-chain automatically. It supports customizable voting-power calculation (tokens stay liquid), submission/voting rules, and several voting systems (single-choice, approval, quadratic). We collected proposals from 59 Ethereum DAOs (minimum 100 proposals and average ≥ 10 votes/proposal) in June 2025, including Aave, Stargate, and Uniswap.

b) Service Nervous System (SNS): The open-source Service Nervous System (SNS)³ provides on-chain DAO governance on the Internet Computer Protocol (ICP) [18], which hosts applications from social networks to fully on-chain AI at low cost. Token holders lock tokens in *neurons* to gain voting and proposal rights; per-DAO parameters (rewards, voting/locking duration, application-specific proposal types) are configurable. We collected proposals from 27 DAOs with > 70 proposals in April 2025 (e.g., OpenChat, ELNA, GoldDAO).

c) OpenGov: Polkadot⁴ [19] is a multi-chain protocol whose parachains handle application-specific tasks (NFTs, identities, bridging) and submit state proofs to the main chain. The on-chain governance system OpenGov decides for the network and individual parachains; any DOT holder can vote. The platform had 1,486 proposals (referenda) in May 2025.

d) DAOstack: DAOstack [20], [21] governed Ethereum DAOs via smart contracts with a native token rewarding contributions and a non-transferable, peer-rated reputation system that assigned voting power. The platform has been inactive since April 2023. We use the dataset collected by Arroyo et al. [22]: 3,572 proposals from 58 DAOs (e.g., Shapeshift, PolkaDAO), with 12,328 votes from 522 participants.

III. CATEGORIZATION METHODOLOGY

To categorize proposals across platforms, we use BERTopic⁵ [23], which captures contextual similarity through transformer-based embeddings rather than relying only on word frequencies. We follow the standard BERTopic architecture where suitable, but replace selected components to better fit our cross-platform setting, as described in Section III-A. We evaluate the results using topic coherence and topic diversity, and compare BERTopic against an LDA baseline in Section III-B.

A. Pipeline

We first standardize and clean proposal text across platforms. We then generate text embeddings, reduce their dimensionality, cluster proposals by semantic similarity, and extract representative keywords. The pipeline is summarized in Figure 1.

Data processing: Since proposal data comes from different platforms, we transform it into a unified format. We then remove markdown fragments, raw URLs, HTML space codes, and trailing whitespace. Some DAO-level information had to be collected manually, the platforms expose different attributes (which limits parts of the cross-platform comparison), and SNS garbage-collection prevented us from querying the full proposal history of some high-volume DAOs; the amount of missing data is limited and is not expected to materially affect the analysis.

Embedding Generation: Each preprocessed proposal is embedded using the transformer `mixedbread-ai/mxbai-embed-large-v1` [24], [25], which ranked among the top clustering models on the MTEB Embedding Leaderboard [26] at the time of our experiments. Each proposal is mapped to a 1024-dimensional vector.

Dimensionality Reduction: To improve clustering efficiency and mitigate the curse of dimensionality [27], we project embeddings to 6–8 principal components per platform using PCA. The number of components per platform was selected so that the resulting clusters remained interpretable

²<https://snapshot.box/>

³<https://dashboard.internetcomputer.org/sns>

⁴<https://polkadot.com/>

⁵<https://bertopic.com/>

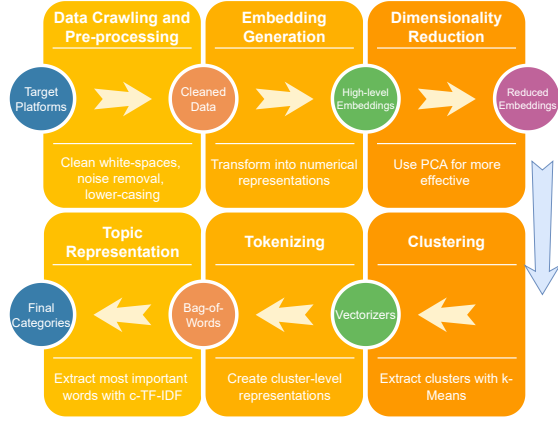


Fig. 1: End-to-end pipeline used for proposal categorization across multiple DAO platforms.

and the topic-coherence scores reported in Section III-B did not drop noticeably.

Clustering: The default BERTopic clusterer, HDBSCAN, classified roughly one third of proposals as outliers and produced 70–100 clusters per platform, making cross-platform comparison impractical. We therefore use k-means clustering. Since the task is unsupervised, we train models with $k \in [5, 12]$ and select the value that best balances topic coherence and topic diversity (Section III-B).

Tokenizing and Topic Representations: After clustering, we vectorize proposals using unigrams and bigrams to preserve expressions such as “smart contract” and “token generation” [23]. We then apply class-based TF-IDF (c-TF-IDF), which treats each cluster as a single document and extracts terms that distinguish it from the rest of the corpus [28], [23]. The top-10 c-TF-IDF terms serve as cluster representatives.

For readability, topic names are assigned by summarizing the highest-weighted c-TF-IDF terms shown in Figures 4d–4b. These labels are used only for interpretation and do not affect the clustering results. All cross-platform comparisons in Section IV rely on the underlying keyword distributions and proposal-level metrics rather than on label semantics.

Inspection of the highest-weighted keywords confirms that the clusters capture distinct proposal types. For example, in SNS (Figure 4d), *Tokenomics & Treasury* is characterized by terms such as *icp*, *token*, *liquidity*, and *treasury*; *Optimization & Security* by *performance* and *security*; *Software Operations* by *execute*, *subnet*, and *release*; and *Repositories & Dev. Workflow* by *git*, *commit*, and *changelog*. Each cluster’s top keywords name a distinct functional domain — token economics, infrastructure security, runtime operations, source-control hygiene — without much keyword overlap across clusters, supporting the interpretation that the clustering recovers substantively distinct topics.

a) *Unified vs. Platform-Specific Topic Modeling:* We initially tested a unified topic model across all platforms, using subsampling to avoid dominance by larger datasets. However, the resulting topics largely separated platforms rather than identifying shared governance themes, because platform-

specific terminology such as SNS canisters, DAOstack relayers, or OpenGov referenda dominated the clusters.

We therefore train separate topic models for each platform. This better captures platform-specific governance patterns and allows shared themes to emerge through cross-platform comparison. When similar topics appear independently across platform-specific models, they provide stronger evidence of common governance themes.

B. Model Selection Evaluation

We evaluate model quality using the top-10 topic keywords.

Topic Coherence Score (TC): Topic coherence measures whether words within a topic tend to co-occur in similar contexts [29]. We use the c_v coherence measure introduced by Röder *et al.* [30], which performed well in human topic-ranking evaluations.

Topic Diversity Score (TD): Topic diversity is the number of unique words across all topics divided by the total number of topic words. Repeated words across topics reduce the score. While coherence captures semantic consistency within topics, diversity captures separation between topics.

BERTopic and LDA comparison: Since LDA is widely used in prior DAO governance work, we include it as a baseline. For both BERTopic and LDA, we use the same preprocessing pipeline. For each method and each number of topics from 5 to 12, we train 25 independent models with different random seeds and report mean coherence and diversity scores.

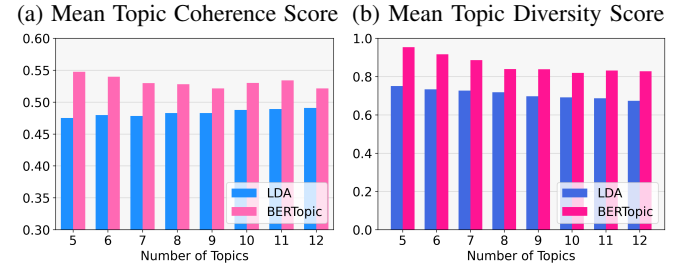


Fig. 2: Training with BERTopic consistently produces more coherent and more diverse topics than LDA across all tested numbers of topics for the dataset.

Figure 2a shows that topic coherence remains relatively stable as the number of topics increases. LDA achieves average TC scores around 0.47–0.49, whereas BERTopic reaches 0.52–0.55. Figure 2b shows that topic diversity declines as the number of topics increases for both approaches, but BERTopic remains higher, with scores between 0.82 and 0.95 compared to 0.67–0.75 for LDA.

Overall, BERTopic produces more coherent and diverse topics and was therefore chosen for the cross-platform analysis.

IV. CROSS-PLATFORM COMPARISON

We compare governance dynamics across the four platforms along six dimensions: proposal categories and their distribution, approval and rejection rates, proposer diversity, temporal trends in activity, decision duration, and the relationship

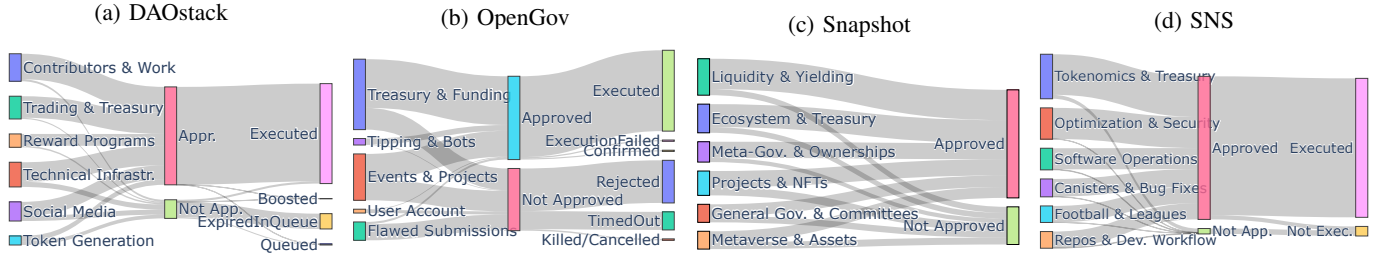


Fig. 3: Sankey diagrams showing proposal topics and lifecycle. SNS shows the highest approval rate (96.3%), while Snapshot and OpenGov display more critical voter behavior and lower acceptance levels. The topics with the highest rejection rates differ by platform; technical and operational topics are most consistently approved.

between proposal length and rejection. To our knowledge, this is the first cross-platform, proposal-level analysis linking proposal content to several governance dimensions.

Main proposal categories and community priorities:

Using the NLP pipeline from Section III-A, we extracted five to six topics per platform by maximizing topic coherence and diversity. Figures 4d–4b show the most important c-TF-IDF keywords per topic, which were used to assign interpretable topic labels. Figure 3 summarizes proposal lifecycles and approval outcomes, while Figure 5 visualizes the resulting topic clusters in two-dimensional PCA space.

Financial topics dominate across all platforms, highlighting the central role of treasury management, funding, and liquidity in DAO governance. In DAOstack, Trading & Treasury, Farming and Reward Programs, and Token Generation account for almost 40% of proposals. In OpenGov, nearly 50% of proposals concern Treasury & Funding, while Snapshot’s Liquidity & Yielding and Ecosystem & Treasury topics jointly cover roughly 45%. In SNS, around 30% of proposals fall into the Tokenomics & Treasury topic.

Beyond finance, the platforms differ substantially. SNS and DAOstack contain many technical and infrastructure-oriented proposals, accounting for roughly 47% and 21% of proposals, respectively. This is particularly pronounced in SNS, where smart-contract upgrades are governed directly through proposals. Snapshot, in contrast, contains no clearly technical topic; its non-financial proposals mostly concern general governance, projects and NFTs, and metaverse-related assets. OpenGov shows lower topic diversity overall, likely because it governs the Polkadot network as a whole rather than many independent DAOs. Finally, SNS contains a domain-specific Football & Leagues topic, driven by one DAO, illustrating how DAO governance can also support highly specialized communities.

Overall, proposal distributions show that financial decisions dominate DAO governance, while technical, community-oriented, and domain-specific topics reveal how governance priorities vary across platforms.

To assess topic quality, we computed silhouette scores [31] (Euclidean distance, in the reduced 6–8 dimensional PCA space used for clustering). Scores ranged from 0.025 for Snapshot to 0.13 for SNS. The absolute values are low: Snapshot’s 0.025 in particular indicates substantial cluster overlap, consistent with the broader thematic spread of off-

chain governance proposals on that platform. We therefore do not treat the silhouette scores as standalone evidence of meaningful categories. Stronger evidence comes from (i) the interpretable c-TF-IDF keyword profiles in Figures 4d–4b, which separate the clusters along thematically coherent axes, and (ii) the approval-outcome associations in Table II. The triangulation of these three signals supports the claim that the extracted clusters capture substantively distinct proposal types rather than artifacts of a single projection or metric.

Platform	χ^2 (df)	p	Cramér’s V
DAOstack	262.4 (5)	< .001	0.27
OpenGov	466.7 (4)	< .001	0.56
Snapshot	1189.4 (5)	< .001	0.25
SNS	366.5 (5)	< .001	0.20

TABLE II: Chi-square statistics and Cramér’s V per platform; all associations are significant at $p < .001$.

Approval and rejection rates: Approval rates differ strongly across platforms and topics, reflecting differences in governance design, proposal content, and community behavior. SNS has the highest overall approval rate at 96.3%, while Snapshot and especially OpenGov exhibit more critical voting behavior, with approval rates of 76.5% and 57.1%.

Within platforms, topic also matters. DAOstack has generally high approval rates, but Token Generation & Testing reaches rejection rates of up to 40%, likely due to experimental or underdeveloped submissions. In Snapshot, more subjective topics such as metaverse assets and NFTs are rejected more often. OpenGov’s high rejection rate is shaped by two factors: it contains several automated, duplicate, or flawed submissions, and its mechanism for editing proposal descriptions is often used to actively encourage “No” votes in order to close unviable proposals faster. I.e., a rejection there does not always indicate community disapproval per se. In SNS, Tokenomics & Treasury has the highest rejection rate, just below 10%, while all other topics remain below 4%. This is consistent with SNS treating treasury-related proposals as critical and requiring stronger approval thresholds. Across platforms, technical and operational proposals are consistently the most likely to be approved; the highest-rejection topics vary by platform.

For the chi-square analysis, we define approval as the outcome of the community vote itself: a proposal is “approved” if it received a majority of “Yes” votes among votes cast,

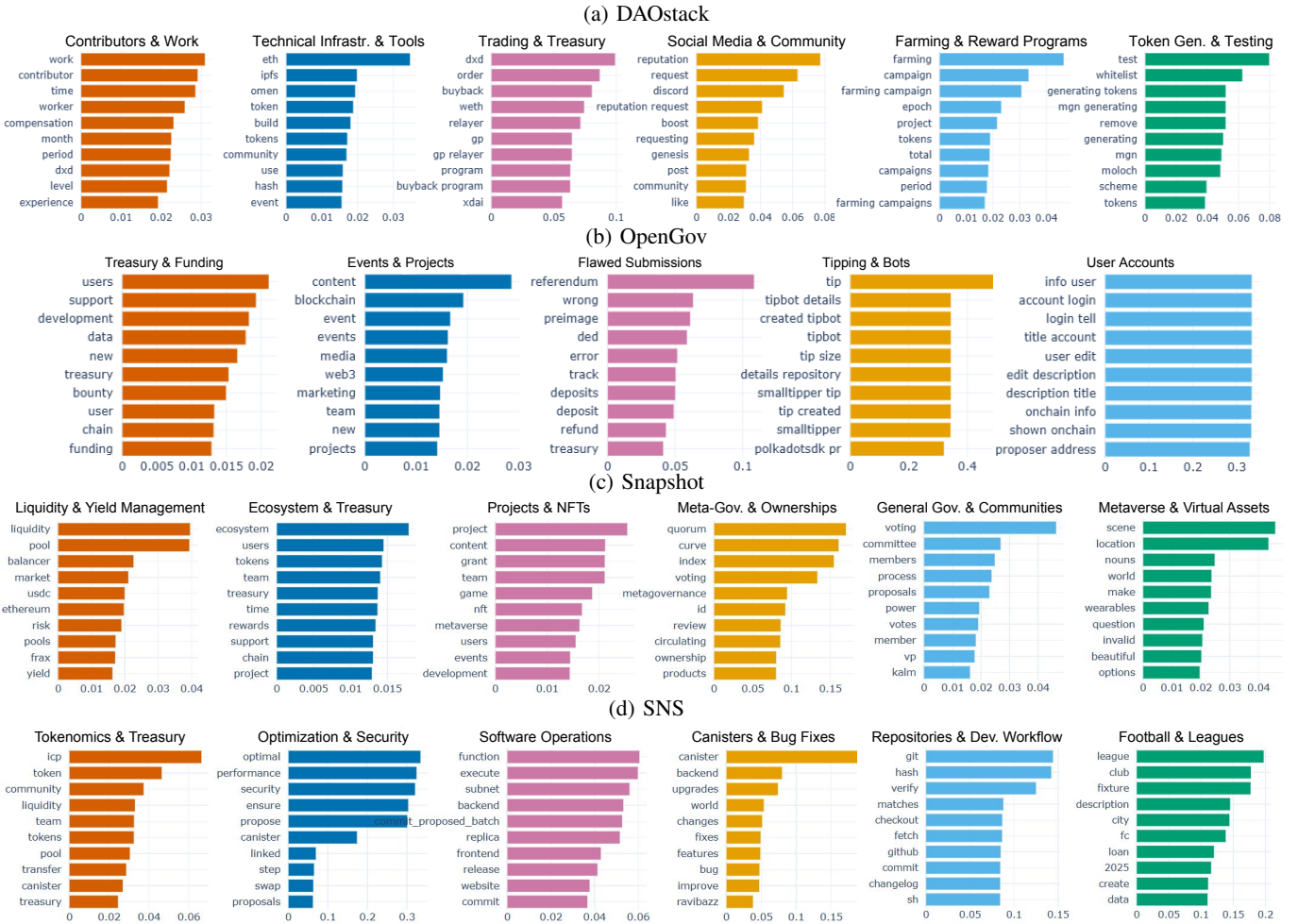


Fig. 4: c-TF-IDF scores for the topics identified per platform (ordered by topic frequency), illustrating diversity of priorities.

and “not approved” otherwise, computed identically across all four platforms and matching the definition used in the proposal-length analysis (Table III). This measure is independent of what happens after the vote. Downstream execution outcomes—e.g., DAOstack proposals executed despite lacking majority approval (likely a smart-contract or threshold-related bug), OpenGov proposals timed out for missing decision deposits or killed/cancelled in cases of malicious submissions, and SNS proposals with majority “Yes” support that were nonetheless not executed because they failed to clear the required voting-power participation threshold—reflect platform-specific execution mechanics rather than voter sentiment, and are reported separately in Figure 3 but do not enter the approval variable used in Table II.

We test the relationship between proposal topic and approval outcome using chi-square tests and Cramér’s V , a normalized effect-size that ranges from 0 (no association) to 1 (perfect dependence) and is comparable across contingency tables of different sizes. As shown in Table II, all platforms exhibit statistically significant associations between topic and approval outcome ($p < .001$). The strength of this relationship varies substantially: SNS shows the weakest association among the

four platforms, consistent with its near-uniform approval rate that caps any topic-driven signal, while OpenGov shows the strongest topic dependence. Cramér’s V (0.20–0.56) quantifies how informative topic alone is about approval outcomes; the strong OpenGov value suggests that proposal content would be a useful input to a predictive model on that platform. Building such a classifier is beyond the scope of this paper but is a natural follow-up.

Proposer diversity: Proposer diversity varies strongly by platform and topic. We measure it as the unique-proposer rate, i.e. the number of distinct proposers divided by the number of proposals in the same topic. OpenGov has the highest diversity, with unique-proposer rates between 49–66%, except for Tipping & Bots at 24%, likely due to automated submissions. SNS shows the strongest concentration: no topic exceeds a 10% unique-proposer rate, and technical topics average around 5%, suggesting dominance by core contributors.

Snapshot and DAOstack show more topic-dependent patterns. In Snapshot, Meta-Governance & Ownerships has a 2.6% unique-proposer rate, while General Governance & Committees and Ecosystem Support & Treasury reach 21–25%. In DAOstack, Social Media has the highest proposer

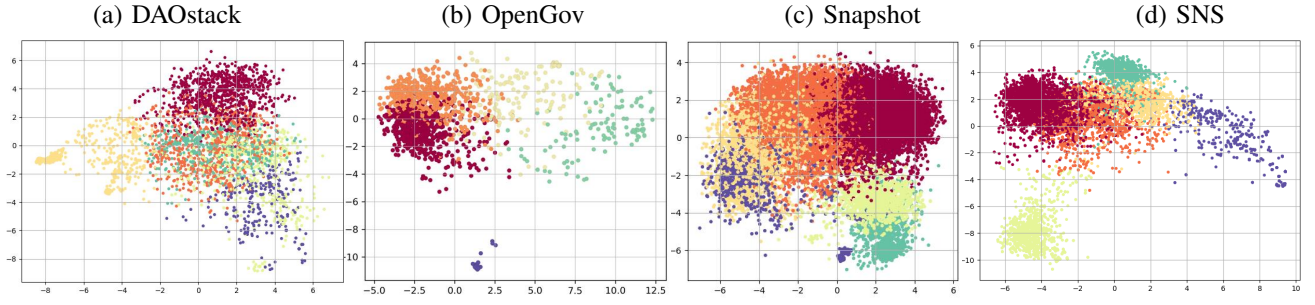


Fig. 5: Proposal-level PCA visualization of the created clusters. Each point is a proposal projected into two dimensions, with colour indicating cluster membership. SNS and OpenGov panels show more visually distinct clusters than the others.

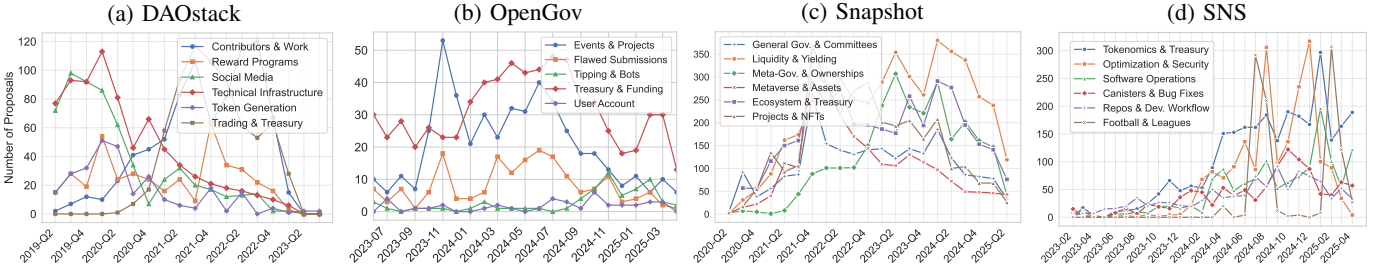


Fig. 6: Quarterly / monthly proposal volumes per topic evolve over time, reflecting changing community priorities. Social media and new project topics peak early on most platforms. DAOstack shows an early surge followed by decline, OpenGov is the most steady platform, Snapshot peaks during the NFT/metaverse hype, and SNS displays significant growth.

diversity at 53.8%, whereas Trading & Treasury and Contributors & Work remain much more concentrated. Overall, proposer diversity is platform-dependent and topic-sensitive; the most concentrated topic on each platform belongs to a different category (Meta-Governance on Snapshot, Trading & Treasury on DAOstack, technical topics on SNS).

Temporal trends in proposal activity: Proposal activity changes substantially over time, as shown in Figure 6. In DAOstack, Snapshot, and OpenGov, the first 12–18 months of activity on each platform often centre on projects, events, and social-media-related proposals, likely reflecting efforts to attract users and build community. Over time, attention shifts toward infrastructure, governance, and long-term development.

DAOstack shows strong early momentum in 2019 and 2020, followed by a steady decline until the platform became inactive in 2023. High gas fees and missing governance features such as vote delegation may have contributed to this decline [32]. Snapshot exhibits high activity from 2021 to early 2024, coinciding with the NFT and metaverse boom, followed by a decline after Q3 2024. SNS, in contrast, grows strongly in 2024, while OpenGov maintains comparatively steady activity with periodic topic-specific peaks.

Decision duration: Proposal duration varies widely across platforms, from less than one day on SNS to several weeks on DAOstack and OpenGov. Within each platform, financial/treasury and project-related proposals take the longest to resolve, while technical and operational topics are decided fastest (e.g., in DAOstack, Token Generation averages ≈ 24 days versus ≈ 11 days for Technical Infrastructure; in

SNS, Tokenomics & Treasury ≈ 1.7 days versus ≈ 0.2 – 0.6 days for the technical and operational topics (e.g., Optimization & Security, Software Operations)), suggesting longer deliberation for resource-allocation and strategic decisions.

Proposal length and rejection: Figure 7 shows that OpenGov and Snapshot proposals are substantially longer than DAOstack and SNS proposals. Across platforms, financial and project- or event-related topics are consistently among the longest, suggesting that resource-allocation and strategic decisions require more detailed justification; the absolute longest topic varies (e.g. Contributors & Work on DAOstack, Events & Projects on OpenGov). Technical topics tend to be shorter.

To study the relationship between proposal length and rejection, we group Snapshot and OpenGov proposals into short (≤ 150 words), medium-length (151–550 words), and long (550+ words) proposals. Since SNS proposals are much shorter on average, we use separate thresholds: short (0–40 words), medium-length (41–200 words), and long (200+ words). Proposals receiving more than 50% and 75% “No” votes are labeled as rejected and strongly rejected, respectively. DAOstack is omitted from this analysis because the required metadata is unavailable.

Snapshot and OpenGov show higher rejection rates for short proposals: 26.3% for Snapshot and 48.8% for OpenGov. This may reflect insufficient information for voters to evaluate short submissions. SNS shows the opposite pattern, with rejection rates increasing for longer proposals. Part of this effect is structural rather than behavioural: SNS classifies treasury and other consequential proposals as “critical” and applies a higher

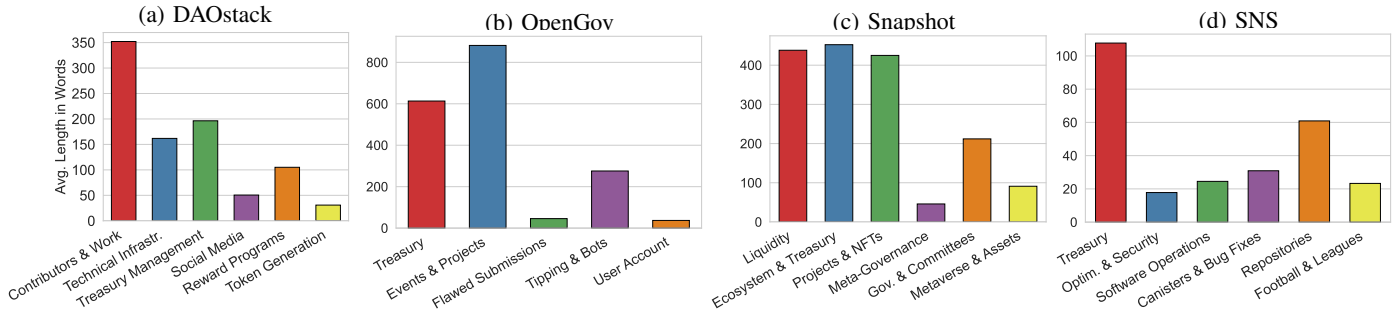


Fig. 7: Mean proposal length per topic (**note differing Y-axis scales**). Proposals on OpenGov and Snapshot are typically several times longer than those on DAOstack and SNS; financial and project-related topics are among the longest everywhere.

Snapshot: shorter proposals are more likely to be rejected.

Proposal Type	Total	Rejected	Strongly Rejected
Short (0–150 words)	8,868	2,333 (26.3%)	1,701 (19.2%)
Medium (151–550)	6,296	1,395 (22.2%)	1,018 (16.2%)
Long (550+)	3,538	676 (19.1%)	484 (13.7%)

OpenGov: shorter proposals are more likely to be rejected.

Proposal Type	Total	Rejected	Strongly Rejected
Short (0–150 words)	521	254 (48.8%)	229 (44.0%)
Medium (151–550)	517	188 (36.4%)	132 (25.5%)
Long (550+)	446	195 (43.7%)	150 (33.6%)

SNS: longer proposals are more likely to be rejected.

Proposal Type	Total	Rejected	Strongly Rejected
Short (0–40 words)	6,024	108 (1.8%)	62 (1.0%)
Medium (41–200)	2,612	179 (6.9%)	116 (4.4%)
Long (200+)	422	44 (10.4%)	24 (5.7%)

TABLE III: Number and percentage of proposals rejected (>50% “No” votes) and strongly rejected (>75% “No” votes), grouped by proposal length and platform.

participation threshold (20% of voting power) than the 3% required for ordinary proposals. In total, 583 SNS proposals were not executed: 331 of these failed to receive a majority of “Yes” votes (consistent with Table III), while the remaining 252 received a “Yes” majority ($\approx 93\%$ “Yes” on average) but did not meet the participation threshold. Overall, successful proposals appear to balance clarity with sufficient detail.

Attribute co-variation: Several proposal-level attributes co-vary with topic identity. Topic is significantly associated with approval (Table II). Financial topics have the longest decision times and consistently rank among the longest proposals, though the absolute longest topic varies. The length–rejection relationship is platform-specific: monotonically decreasing on Snapshot, monotonically increasing on SNS (partly via the critical-proposal quorum), and non-monotonic on OpenGov. Proposer concentration is also topic-sensitive, with the most concentrated topic differing across platforms.

V. RELATED WORK

DAO governance has recently attracted substantial research interest, especially around participation, centralization, and voting dynamics. Table IV summarizes closely related studies.

Wang et al. [14] conduct a broad empirical study of Snapshot DAOs, classifying proposals into ten categories and

analyzing voting rate, duration, language, and power concentration; they emphasize diverse governance contexts and persistent centralization risks. Pena-Calvin et al. [15] focus on the Decentraland DAO and use topic modelling to show that community, content-creation, governance, and voting-process topics dominate, while documenting low turnout and significant centralization. Mosley et al. [16] study Dash and report rising proposal activity but declining participation, raising fairness concerns about concentrated voting power. Okutan et al. [10] analyze SNS DAOs and find high engagement, low costs, sustained participation, and fast on-chain decisions, in contrast to declines reported for other frameworks. Kumar et al. [33] apply LDA to DAOstack and Aragon proposals to identify investment, community, and technical themes, but do not link categories to voting outcomes. In contrast, this paper provides a large-scale cross-platform analysis using transformer-based topic modelling (with an LDA baseline) that links platform-specific categories to approval rates, proposer diversity, length, and temporal dynamics.

VI. CONCLUSION

This paper presented a large-scale analysis of over 32,000 proposals from four DAO ecosystems: Snapshot, SNS, OpenGov, and DAOstack. Using a transformer-based topic modeling pipeline, we categorized proposals by content and analyzed how proposal topics relate to governance outcomes, proposer diversity, proposal length, and temporal activity patterns.

Our results show that proposal content is strongly linked to DAO governance dynamics. Financial and treasury-related proposals dominate activity and have the longest decision times across platforms, consistent with the higher stakes of collective resource allocation; the highest-rejection topics, however, are platform-specific. Technical and operational proposals are typically shorter, faster to resolve, and more frequently approved, consistent with voter deference on specialized matters where proposers tend to have more domain knowledge. Proposer diversity is highly topic-dependent: aggregate participation measures can hide substantial concentration in specific decision domains.

These findings suggest that *what* is being voted on is as important as *how* voting is conducted. Governance systems could therefore benefit from surfacing proposal categories,

Study (Platforms)	[14] (Snapshot)	[15] (Decentraland)	[33] (DAOstack, Aragon)	[16] (Dash)	[10] (SNS)	This paper (OpenGov, SNS, Snapshot, DAOstack)
Scope (DAOs/Prop.)	581 / 16,246	1 / 2,547	— / >3,000	1 / 550	14 / >3,000	144 / 32,819
Method (# categories)	k-Means (10)	k-Means, LDA (10)	LDA (6)	LDA (6)	pre-defined SNS types (3)	BERTopic + LDA (5–6 per platform)
Key insight	Diverse governance, centralization risks	Low turnout, centralization, manipulation	Frequent terms dominate, topic overlap	Declining participation, centralization	High engagement, fast decisions	Governance patterns evolve and vary across platforms, length and topic significantly associated with approval

TABLE IV: Comparison of related cross-platform and topic-modelling studies on DAO governance. Our work covers more DAOs and proposals than the listed prior studies, combines BERTopic with an LDA baseline, and jointly analyzes content categories with approval outcomes, proposer diversity, proposal length, and temporal trends.

supporting proposers in choosing an appropriate level of detail, and adapting voting thresholds or review processes to the criticality of different proposal types.

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